1. I/O Interface

- The I/O interface acts as a bridge between the CPU and peripheral devices.
- It includes **data registers**, **status registers**, and **control registers** to manage communication.
- The interface ensures proper synchronization between the processor and I/O devices.

2. Types of I/O Operations

- **Programmed I/O**:
  - The CPU actively monitors the status of the I/O device and performs data transfer.
  - It is simple but inefficient, as the CPU is busy waiting for the device to be ready.
- **Interrupt-Driven I/O**:
  - The I/O device interrupts the CPU when it is ready for data transfer.
  - This allows the CPU to perform other tasks while waiting for the I/O device.
- **Direct Memory Access (DMA)**:
  - A dedicated DMA controller handles data transfer directly between memory and the I/O device, bypassing the CPU.
  - This is highly efficient for large data transfers.

3. Memory-Mapped I/O vs. Isolated I/O

- **Memory-Mapped I/O**:
  - I/O devices share the same address space as memory.
  - The same instructions used for memory access can be used for I/O operations.
- **Isolated I/O**:
  - I/O devices have a separate address space.
  - Special instructions are required for I/O operations.

4. Interrupts

- Interrupts are signals sent by I/O devices to the CPU to indicate that they need attention.
- Types of interrupts:
  - **Hardware Interrupts**: Triggered by external devices.
  - **Software Interrupts**: Triggered by programs.
- The CPU uses an **Interrupt Service Routine (ISR)** to handle interrupts.

5. Bus Communication

- The I/O devices communicate with the CPU and memory via a **system bus**.
- The bus includes:
  - **Data lines** for transferring data.
  - **Address lines** for identifying devices or memory locations.
  - **Control lines** for managing operations.

Peripheral Devices in Computer Science

12

A **peripheral device** is any external hardware component connected to a computer that enhances its functionality but is not part of its core system. These devices are primarily used for **input**, **output**, **storage**, or **communication** purposes. While a computer can operate without peripherals, they significantly improve usability and expand the system's capabilities.

Types of Peripheral Devices

Peripheral devices are broadly categorized into **input devices**, **output devices**, and **input/output devices**:

- **Input Devices**: These allow users to send data or commands to the computer. Examples include keyboards, mice, scanners, webcams, and microphones.
- **Output Devices**: These display or present processed data to the user. Examples include monitors, printers, speakers, and projectors.
- **Input/Output Devices**: These perform both input and output functions. Examples include hard drives, USB drives, memory cards, and network interface cards (NICs).

Examples and Functions

- **Keyboard**: Enables users to input text and commands.
- **Mouse**: Provides point-and-click interaction with the system.
- **Monitor**: Displays visual output, often referred to as a "soft copy."
- **Printer**: Produces physical copies of documents or images, known as "hard copies."
- **Hard Drive**: Stores data and allows reading and writing operations.
- **USB Drive**: Facilitates portable data storage and transfer.
- **NIC**: Connects a computer to a network for data exchange.

Importance of Peripheral Devices

Peripheral devices are crucial for enhancing a computer's functionality. They enable user interaction, data storage, communication, and efficient task execution. For instance, input devices like keyboards and mice allow users to control the system, while output devices like monitors and printers present processed data. Storage devices ensure data is saved and accessible, and communication devices enable network connectivity.

Connection Methods

Peripheral devices connect to computers through various interfaces, including USB, Bluetooth, Wi-Fi, HDMI, Ethernet, and Thunderbolt. Each method offers different speeds and compatibility, catering to specific device requirements.
Peripheral devices play a vital role in making computers versatile and user-friendly, supporting a wide range of tasks and applications.

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